

GARRF VIRTUAL STRUCTURAL HEALTH MONITORING LABORATORY

STUDENT HELP & EXPERIMENT GUIDE

Piezoelectric (PZT) Based EMI & Guided Wave Structural Health Monitoring

Developed by

Dr. Annamdas Venu Gopal Madhav &

Annamdas Shantanu Vasudev Krishna

Gopalkrishna Advanced Rural Research Foundation (GARRF)

WELCOME TO THE GARRF VIRTUAL LABORATORY

This virtual laboratory has been designed to let you:

Explore → Experiment → Observe → Compare → Analyze → Understand

You are not expected to understand everything before starting.

Try the controls. Change the parameters. Watch the waves. Observe the waveform. Study the FFT. Introduce simulated damage. Compare the results.

The purpose is to help you develop an intuitive understanding of **Piezoelectric (PZT) based Structural Health Monitoring (SHM)**.

1. WHY THIS VIRTUAL LABORATORY?

This type of Structural Health Monitoring laboratory is commonly available in leading international universities and advanced research laboratories.

A few IITs and premier research institutes in India are actively working in the area of **Piezoelectric (PZT) based Structural Health Monitoring**.

The **Gopalkrishna Advanced Rural Research Foundation (GARRF)** is bringing this type of advanced learning to rural students **free of cost**, so that geography, financial limitations and lack of access to expensive laboratory equipment do not become barriers to engineering education.

The objective is to give students an opportunity to explore concepts that they may otherwise encounter only in advanced university laboratories or research institutions.

2. IMPORTANT NOTE ABOUT THE VIRTUAL BEAM AND PLATE

The **Beam** and **Plate** in this laboratory are **generic educational virtual structures**.

They do not currently represent a particular real-world beam or plate with a complete set of physical properties.

For example, the present generic model does not define a specific:

- Length
- Width
- Thickness
- Material grade
- Young's modulus
- Poisson's ratio
- Density
- Damping
- Boundary condition
- PZT size
- PZT material specification
- Adhesive properties
- Real crack geometry

Therefore:

The Beam and Plate are provided for conceptual understanding and experimentation, not as calibrated models of a particular real engineering structure.

3. WHY ARE THE PROPERTIES IMPORTANT?

In a real engineering experiment, the response depends on the actual physical characteristics of the structure and sensor.

Changing the:

- Geometry
- Material
- Thickness

- Boundary conditions
- PZT properties
- PZT location
- Adhesive layer
- Excitation
- Environmental conditions

can change the measured response.

Consequently:

Actual results from a real beam or plate will be different from the results generated by this generic virtual laboratory.

The numerical values, waveforms, FFT peaks and damage indicators from this simulation should therefore **not be treated as universal engineering values**.

4. WHAT IS THE MAIN THING YOU SHOULD LEARN?

The most important concept is the complete SHM chain:

PZT Excitation



Structural Waves



Wave Propagation



Interaction with Damage



Reflection / Scattering / Attenuation



PZT Sensing



Signal Acquisition



Time-Domain Analysis



FFT / Frequency-Domain Analysis



Damage Identification

This is the fundamental idea behind the experiment.

PART 1 — SET UP YOUR EXPERIMENT

Step 1 — Select the Structure

Choose:

Beam

or

Plate

Remember that these are generic educational structures.

Step 2 — Position the PZT

Drag the PZT patch to the desired location.

Observe its position relative to the structure.

For a controlled Healthy-versus-Damaged experiment, keep the PZT position unchanged.

Later, move the PZT and conduct another experiment to see how sensor position affects the response.

Step 3 — Select Excitation Frequency

Choose an excitation frequency.

Different frequencies can interact differently with structural modes and damage conditions.

You do not need to understand this completely before experimenting.

Try different frequencies and observe what happens to the FFT.

Step 4 — Select Excitation Type

You can select:

SINE

Single-frequency excitation.

Useful for studying frequency-specific response.

SWEEP

Changing frequency excitation.

Useful for exploring a broader frequency response.

PULSE

Short excitation.

Useful for understanding wave propagation, reflections and time-domain behavior.

PART 2 — SELECT MONITORING DURATION

You can choose:

1 MINUTE

2 MINUTES

or

5 MINUTES

For your first experiment, choose:

1 MINUTE

Later, try 2 minutes and 5 minutes.

IMPORTANT — THE TIMER IS DELIBERATELY EXPLORATORY

The monitoring-duration options are provided to **guide your experiment**, but the current learning version deliberately allows the experiment to continue rather than forcing an automatic stop exactly at the selected time.

This is intentional.

The idea is to encourage you to:

observe → discover → explore

rather than simply wait for the software to terminate the experiment.

Therefore:

If you select 1 minute and the experiment continues, don't worry.

Monitor for approximately 1 minute and press:

STOP

manually.

The same applies to 2 minutes and 5 minutes.

Example

1 minute selected

→ Observe for approximately 1 minute

→ Press STOP if it is still running

Or

2 minutes selected

→ Observe for approximately 2 minutes

→ Press STOP if it is still running

Or

5 minutes selected

→ Observe for approximately 5 minutes

→ Press STOP if it is still running

There is nothing wrong if you need to press STOP manually.

PART 3 — START WITH HEALTHY

Always begin with:

HEALTHY

This establishes your reference condition.

It is your:

Healthy Baseline or

Stage 0

Healthy Experiment

1. Select **Healthy**.
 2. Confirm that Severity is locked.
 3. Select the monitoring duration.
 4. Press **START**.
 5. Watch the structure.
 6. Watch the simulated waves.
 7. Watch the live waveform.
 8. Watch the FFT.
 9. Observe for your selected duration.
 10. Press **STOP** manually if necessary.
-

5. WHY IS SEVERITY LOCKED WHEN HEALTHY IS SELECTED?

This is intentional.

When the structure is:

HEALTHY

there is no simulated damage.

Therefore there is no damage severity to specify.

The interface should show:

Severity: N/A — Healthy

and keep the control locked.

When you select a damage condition, Severity becomes active.

PART 4 — INTRODUCE SIMULATED DAMAGE

After exploring Healthy, select one of the available damage conditions.

Possible conditions include:

- **Crack**

- **Debonding**
 - **Loosened Connection**
 - **Local Stiffness Reduction**
-

CRACK

Represents a localized discontinuity in the virtual structure.

It can change simulated wave propagation, reflection and scattering.

DEBONDING

Represents a simulated weakening or loss of the PZT/structure interface response.

This is important because sensor bonding can also influence PZT measurements in real experiments.

LOOSE CONNECTION

Represents a simulated change in structural connection or boundary stiffness.

LOCAL STIFFNESS REDUCTION

Represents a simulated localized reduction in structural stiffness.

PART 5 — UNDERSTANDING SIMULATION SEVERITY

When you select a damage condition, Severity becomes active.

For example:

Crack — Severity 30%

or

Crack — Severity 60%

Use Severity to explore how changing the simulated damage parameter affects the response.

IMPORTANT

Severity is a **simulation parameter**.

It does not automatically mean:

- 30% crack length
- 30% structural failure
- 30% material failure
- 30% loss of strength

unless a validated physical model specifically establishes that relationship.

Therefore, think of it as:

Simulation Severity

PART 6 — RUN THE DAMAGED EXPERIMENT

For your first comparison, use:

Healthy → Crack

Keep the following unchanged:

- Structure
- PZT position
- Frequency
- Excitation type
- Monitoring duration

Change only:

Healthy → Crack

Then:

1. Select **Crack**.
2. Set the Simulation Severity.
3. Press **START**.
4. Watch the structure.
5. Watch the waves.
6. Watch the waveform.
7. Watch the FFT.

8. Observe for your selected duration.
9. Press **STOP** manually if necessary.

You now have:

Healthy

versus

Simulated Crack

PART 7 — UNDERSTAND THE WAVE VISUALIZATION

The moving wave represents the simulated mechanical disturbance travelling through the virtual structure.

When the wave encounters a simulated structural change, you may observe:

- Reflection
- Scattering
- Attenuation
- Local vibration changes
- Changes in the wave pattern

The objective is to develop an intuitive understanding of how a structural change can influence wave propagation.

PART 8 — UNDERSTAND THE LIVE WAVEFORM

The waveform represents the simulated PZT response as a function of time.

Compare:

Healthy waveform

with

Damaged waveform

Look for changes in:

- Amplitude
- Shape
- Peaks

- Periodicity
- Energy
- Arrival time
- Reflections

Remember:

A difference in a signal does not automatically prove physical damage.

PART 9 — UNDERSTAND THE FFT

FFT converts the time-domain signal into a frequency-domain representation.

Look for:

- Peak frequency
- Peak amplitude
- New peaks
- Reduced peaks
- Peak shifts
- Changes in spectral shape

Damage may change the frequency-domain response because the structural mechanical response has changed.

PART 10 — WHY DOES THE PZT RESPONSE CHANGE?

The basic physical process is:

PZT



Excites the structure



Mechanical wave travels



Wave encounters a structural condition

↓

Wave is reflected / scattered / attenuated

↓

Changed response reaches the PZT

↓

PZT produces a changed electrical response

Therefore:

A change in the measured electrical response can contain information about a change in structural condition.

This is one of the fundamental concepts behind PZT-based SHM.

PART 11 — HEALTHY VS DAMAGED COMPARISON

Ask yourself:

Wave pattern

Did the simulated propagation change?

Amplitude

Did the response become stronger or weaker?

Waveform

Did the time-domain shape change?

FFT

Did a frequency peak move or change amplitude?

RMS

Did the overall signal magnitude change?

Correlation

How similar are the two signals?

Damage Indicator

Did the laboratory-defined indicator show a larger difference?

PART 12 — CHANGE ONE VARIABLE AT A TIME

This is very important.

If you want to study the effect of a crack:

Keep constant:

Beam

PZT position

Frequency

Sine

1 minute

Change only:

Healthy → Crack

This makes the comparison easier to understand.

Once you understand the basic experiment, deliberately change other parameters and explore their effects.

PART 13 — EXPLORE DIFFERENT EXPERIMENTS

Try:

Experiment 1

Healthy → Crack

Experiment 2

Crack Severity 20% → Crack Severity 60%

Experiment 3

Sine → Sweep

Experiment 4

Pulse excitation

Experiment 5

Change frequency

Experiment 6

Move the PZT

Experiment 7

Healthy → Debonding

Experiment 8

Healthy → Loosened Connection

Experiment 9

Healthy → Local Stiffness Reduction

The purpose is to discover how different conditions influence the simulated response.

PART 14 — EXPLORE DIFFERENT MONITORING TIMES

Try:

1 minute

then:

2 minutes

then:

5 minutes

Use the additional time to observe:

- Signal stability
- Repeating patterns
- Wave behavior
- Waveform behavior
- FFT behavior
- Changes during continued excitation

Again:

If the experiment does not stop automatically, don't worry.

Press **STOP** manually after the intended monitoring period.

PART 15 — IF YOU PRESS STOP

Manual STOP is allowed.

If the experiment is still running after your intended monitoring period:

Press STOP.

You have not done anything wrong.

For a controlled comparison, try to use approximately the same observation duration for Healthy and Damaged conditions.

For example:

Healthy → 1 minute → STOP

↓

Crack → 1 minute → STOP

↓

Compare

PART 16 — CONNECTING YOUR EXPERIMENT TO ARJUN

After you have completed your virtual experiment, you can use:

CONNECT TO ARJUN

The GARRF Virtual Laboratory prepares the experiment information in a **JSON data file** for ARJUN.

What is JSON?

JSON (JavaScript Object Notation) is a machine-readable data format.

You do **not** need to understand or edit JSON.

It allows the virtual laboratory to organize the experiment information so that an AI system such as **ARJUN** can understand the experiment parameters and results.

The JSON can contain information such as:

- Structure type
- PZT position
- Excitation frequency
- Excitation type
- Monitoring duration

- Healthy / Damaged condition
 - Damage type
 - Simulation Severity
 - Waveform information
 - FFT information
 - Calculated indicators
 - Experiment information
-

DO I NEED TO SAVE THE JSON FILE?

If you press CONNECT TO ARJUN

Normally, you do not need to manually save the JSON file first.

The laboratory prepares the experiment data for the ARJUN connection.

Follow the instructions presented by the laboratory/ARJUN interface.

The purpose is to make the transition from:

Virtual Experiment → AI Analysis

as simple as possible.

WHY IS THE JSON FILE USED?

The JSON file is essentially the **machine-readable experiment record**.

Think of it as:

Your virtual laboratory experiment written in a format that software and AI can understand.

You do not have to read the JSON yourself.

ARJUN can use the structured information to understand:

What experiment did you perform?

What structure did you select?

What excitation did you use?

What damage condition did you introduce?

What severity did you select?

What response was generated?

This allows ARJUN to continue the analysis from the simulation.

PART 17 — CAN I STILL DOWNLOAD OR SAVE THE DATA?

Yes.

If the laboratory provides a **Download Experiment Data** option, you can save the experiment data to your computer.

This is useful when you want to:

- Keep a record of your experiment
- Repeat the analysis later
- Share the experiment with a teacher
- Compare different experiments
- Keep your own research record
- Upload the data manually to ARJUN later

The JSON file should normally be kept **unchanged**.

Do not manually edit the JSON unless you know exactly what you are doing.

PART 18 — CAN I USE CSV FILES WITH ARJUN?

Yes.

ARJUN is not limited to the GARRF Virtual Laboratory JSON files.

If you have experimental or simulated data in another format, you may be able to upload supported files such as:

- **CSV**
- **Excel**
- **PDF**
- **Images / screenshots**
- **Oscilloscope captures**
- **Other supported experimental data**

For example:

GARRF Virtual Laboratory

Virtual Experiment



CONNECT TO ARJUN



JSON experiment data



ARJUN

Independent experiment

Real laboratory measurement



CSV / Excel / PDF / Image



Upload to ARJUN



ARJUN Analysis

PART 19 — TWO WAYS TO USE ARJUN

METHOD A — FROM THE GARRF VIRTUAL LAB

Complete your experiment.



Press:

CONNECT TO ARJUN



The laboratory prepares the structured experiment information.



ARJUN receives/uses the experiment data.



ARJUN analyzes the simulation.

This is the easiest route for students.

METHOD B — UPLOAD YOUR OWN DATA

You can also open ARJUN independently.

Upload your own:

CSV / Excel / PDF / Image / supported data

Then tell ARJUN what the data represents.

For example:

This is real PZT experimental data from a laboratory beam. Please analyze the waveform and frequency response.

This makes ARJUN useful beyond the GARRF virtual laboratory.

PART 20 — IMPORTANT: SIMULATION DATA VS REAL DATA

Always tell ARJUN where your data came from.

For GARRF Virtual Laboratory data:

This is simulated educational data generated by the GARRF Virtual Structural Health Monitoring Laboratory.

For real laboratory data:

This is real experimental PZT/SHM data collected from a physical specimen.

This distinction is important.

The generic virtual Beam and Plate do not represent a specific physical structure.

Therefore, the simulation results should not be interpreted as measurements from a real beam or plate.

PART 21 — THE COMPLETE GARRF → ARJUN WORKFLOW

The complete learning workflow is:

SELECT STRUCTURE

↓

POSITION PZT

↓

SELECT FREQUENCY

↓

SELECT EXCITATION

↓

HEALTHY

↓

MONITOR

↓

STOP

↓

INTRODUCE DAMAGE

↓

SET SIMULATION SEVERITY

↓

MONITOR

↓

STOP

↓

COMPARE HEALTHY VS DAMAGED

↓

CONNECT TO ARJUN

↓

JSON EXPERIMENT DATA

↓

ARJUN ANALYSIS



UNDERSTAND WHY THE SIGNAL CHANGED

IMPORTANT REMINDER

The JSON file is **not something the student needs to study as programming code.**

It is simply a **machine-readable experiment data file.**

If the GARRF laboratory automatically transfers/prepares it when **CONNECT TO ARJUN** is pressed, the student does not need to separately save the JSON just to make the connection work.

However, downloading and keeping a copy is recommended if the student wants a permanent record of the experiment.

CSV files remain useful and can still be uploaded to ARJUN when working with independently generated or real experimental data.

Making Advanced Structural Health Monitoring Education Accessible to Rural Students

Learn → Explore → Experiment → Observe → Save → Analyze → Understand → Research